SMART DETECTION AND TREATMENT ESTIMATION OF ANTHURIUM DISEASES USING IMAGE BASED AI MODELS

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Project Proposal Report

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DECLARATION

I declare that this is my own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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(Ms Anjalie Gamage)
Signature of the Supervisor

Date.....30/8/25.....

ABSTRACT

Anthurium is valuable ornamental plant because of its commercial and aesthetic importance. Many people have turned to anthurium cultivation because it is low-cost to start, easy to maintain, and can be started in a small space. However, anthurium is susceptible to several fungal, bacterial, and virus diseases, which greatly affect its yield. The biggest problem is infections not only reduce flower quality and yield but also increase the cost of chemical treatments and expert intervention. In this case incurs a huge cost to the cultivators. Specially the small cultivators will face big problem because of the huge cost and it will take lot of time to overcome this issue. So it is necessary to technical system which can give real time, smart and easy diseases detection and management support for Anthurium growers.

This research designs and develop for give a best solution for several anthurium diseases like fungal, virus etc. In this system develop to identify many diseases by integrates computer vision and machine learning techniques to automatically detect diseases after upload Anthurium plant or flower. These diseases detect system has ability to identify not only the type of disease, provide tailored recommendations for resolutions but also can estimate the approximate recovery time based on the infection and environmental condition. This mobile application develops for empower anthurium growers who does not have much agricultural knowledge

The novelty of this system is rare addressed combining automated disease identification with treatment and recovery prediction. This system will be user friendly interface that everyone can easily understand and have easy navigating among interfaces. Hope to develop a mobile app as this is specially developed for cultivators. So, this is support on fields too .

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LIST OF ABBREVIATIONS

Abbreviation	Description
CA	Channel wise Attention
AI	Artificial Intelligent
SE	Squeeze and Excitation
IoT	Internet of Things
ML	Machine Learning
NN	Neural Network
KNN	K-Nearest Neighbor

1.INTRODUCTION

Anthurium is an economically important ornamental plant of significant commercial and aesthetic value. Its culture ‘s heavily affected by diseases caused by fungi, bacteria, and viruses that in addition to yield and quality losses to the producers increase the cost of plant treatments. The aim of this project is to develop a mobile application that utilizes computer vision and machine learning to detect diseases in the field in real-time, suggest treatment and give an indication of the recovery time for Anthurium plants and flowers. The user-friendly interface in the system will allow even the smallholder to cure diseases, cost saving and productivity increasing from his mobile, without needing an extensive knowledge of agriculture.

1.1 Background & Literature Review

Anthurium is a popular ornamental plant. It is also cultivated as a houseplant by many people for its beauty. Both large-scale growers and small-scale growers have challenges to consider in cultivation. These include unpredictable markets, fluctuations, climate changes, soil imbalance and disease outbreaks. Large-scale growers may face a major economic crisis if they do not successfully face the above challenges. So, the main goal of this project is to develop a system that makes it easier to overcome above challenges that Anthurium cultivators face. This background about the, methods and how it effort for the above research.

Among the problems encountered in growing anthuriums, one of the problems that directly affects yield reduction is the occurrence of various diseases. This research will help growers easily identify these diseases, know their treatment methods, and determine the time required for treatment.

If any plant had a disease, it is difficult and complex to recognized what those diseases. So, this reason even experienced agronomists and plant pathologists often fail to identify specific diseases. Not only that but also some time they led to mistaken conclusion and treatment. But later develop a simple but power full system, a mobile application for identity diseases. It could be a valuable

tool for farmers as world lacking psychopathological advices. That system was accurate and time locate. It was really easy for that farmers as the system was a mobile application. So farmers can carry their mobiles when they coming to planting. So that kind of disease detection, mobile system best for detect anthurium diseases. [1]

This anthurium diseases detection mobile app hope to create by basic deep learning tools. The above deep learning tool, CNN is a powerful technique for build complex processes. CNN system created for automated recognition of plant based on its leave images . This CNN can easily recognize disease by both anthurium leave and flower. CNN can train to identify diseases. Established architecture of CNN identified 26 plant diseases among 14 different plants. The success rate of that is 99.35%. This model develops to decease detection and diagnosis using both healthy and diseased plants. This model is also useful early wearing tool. [2]

When talking about datasets and image-based methods that hope to detect Anthurium diseases, this system classify based on hand engineering features , shape based features , texture based features . For this disease detection Use new architecture , combine efficient neural network (CA) for match latency and size of plant . For improve the identification of Anthurium leaf and flower diseases hope to combine squeeze and excitation (SE) block with the backbones. The above best trained model accuracy showed 99.14%. [3]

Anthurium flower cultivation is highly gain income for the cultivators. This Anthurium flower is a one of the most popular and selling flowering plant across in south Asian counties specially in Sri Lanka. But this plant has many diseases like insets . farming these plants economic benefits, multiple risks are associated with cultivating them.But the problem is Manual disease detection methods are expensive and need experts knowledge to overcome this matter.Most diseased plants and flowers are detect by the (IoT) and (AI) technologies .But there is also in a matter as most of them are complex and expensive. In conjunction with these, a sensor model suitable for collecting data on Anthurium leaves and flowers to disease in this research. A Machine Learning (ML) technique based on the latest (NN) . Experiments conducted and comparisons with existing (KNN) techniques showed that the
EFFICIENTNET-B0 technique had shown remarkable performance with 97.70% accuracy [4]

2.1 Research Gap

Most of plant disease detection systems are mainly focus primarily on food crops such as rice, tomato etc with limited research dedicated. But the ornamental plants like Anthurium. By using deep learning models have shown good results in detecting diseases, very few studies have explored the integration of disease identification with practical guidance like treatment recommendations or recovery time estimation. current mobile applications for plant care generally provide either basic cultivation information or simple symptom checkers without the ability to accurately analyze plant and flower images using AI. The lack of a specialized, intelligent, and mobile-based solution for Anthurium disease detection leaves a gap in providing real-time, accessible, and actionable support for farmers and gardeners who cultivate this economically valuable ornamental plant .

Existing System

Research A: Only Detect Diseases of Anthurium plants, flowers

- In research A is only focuses detecting plant diseases from images and no any treatment suggestions or recovery timelines .

Research B: Only Transfer image classification

- CNN-based plant disease detection through transfer learning using ResNet-34 and ResNet-50 models but no treatment recommendation or recovery timeline

Research C: Explainable AI survey without Specific Anthurium or Treatment Recommendation

- Leaf-based plant disease detection techniques and explore explainable AI , but lack any crop-specific system, treatment recommendations, or recovery estimation capabilities

Application / Study	Disease Detection	Treatment Recommendation	Recovery Time Estimation	Web-Based Interface
Research A	✅	❌	❌	❌
Research B	❌	✅ (General)	❌	❌
Research C	❌	❌	❌	❌
▨ **Proposed System**	✅ **(Anthurium-specific)**	✅ **(Automated & Mapped)**	✅ **(AI-based Estimation)**	✅ **(Live & Remote)**

Figure1

2.2 Research Problem

The cultivation of Anthurium is highly susceptible to diseases of plants and flowers, lowering their ornamental quality and diminishing commercial value. Farmers and gardeners frequently cannot distinguish among the diseases, anticipate them early and treat them with delay or improper ways, causing plants to get further damaged. Furthermore, agricultural expertise may not always be easily accessible, for example in rural or resource-poor areas. Service The need Description The development of an Anthurium disease detection system that is Based on computer vision To realize the objective the detection system needs to be The future NTC could be much more efficiently used for cultivar development" Energy (ATEPI) is a crucial energy asset in supporting the long-term growth of domestic economy.

2. Objectives

2.1 Main Objectives

The aim of this research is to introduce more efficient, reliably, and timeless system to anthurium cultivates by using image-based AI models. In this system for case for not only detect disease in

anthurium plant but also provide valuable guidance for recover them. This platform ensure cultivators can prefer this system if they don't have much Knowledge about agriculture. This system specially designs as a user friendly and easy navigation system that anyone can use.

2.2 Special Objectives

- Purpose-Driven Innovation
  Develop Detection of Several Anthurium diseases of both plant and flower through image uploads. Develop Detection system by using CNN.

- Treatment recommend and guiding system
  Develop treatment recommending system for detected diseases and guide for care detected plants and prevent spreading diseases to healthy plants .

- Perfect Decision Support estimating recovery periods to helping growers act without expert intervention.

3.0 Methodology

3.1 System Architecture

User Layer (Mobile App)

- Built with Flutter.
- Allows users to:
- upload images of Anthurium plants and flowers.
- View detection results
- Can get treatment recommendations guide.
- Provide estimated recovery time.

2. Application Layer (Backend Services) • Developed using Python

- Image preprocessing
- By using Keras/Tensor Model inference for disease detection .
- Recovery time estimation based on disease type.
- Knowledge base Treatments guiding

3. AI/ML Layer (Model Processing)

- Training Phase (Offline):
    - Deep learning model trained with Anthurium plant/flower disease datasets.
    - Evaluation and optimization of accuracy.
- Deployment Phase (Online):
    - Optimized models converted to TensorFlow Lite for mobile deployment or served via cloud API.
    - Outputs disease class, severity score, and probability.

4. Data Storage Layer (Cloud & Database)

- Firebase Authentication – user login/management.
- Firebase Storage – stores uploaded images if cloud inference is used.
- Fire store Database – stores detection results, treatment advice, and user history.

5. Output Layer

- Detected disease (plant/flower).
- Recommended treatment plan.
- Estimated recovery time.
- Disease tracking history & location map

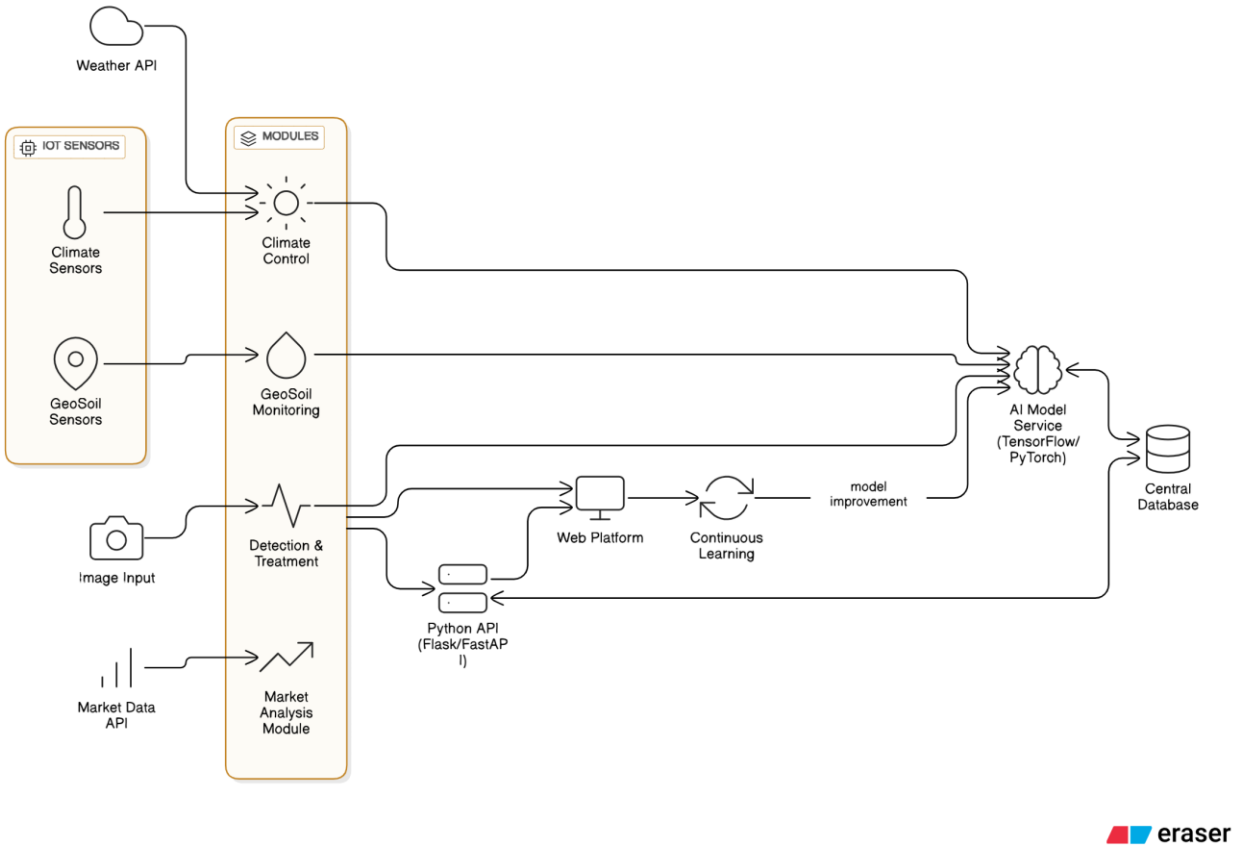


Figure2 : System Architecture Diagram

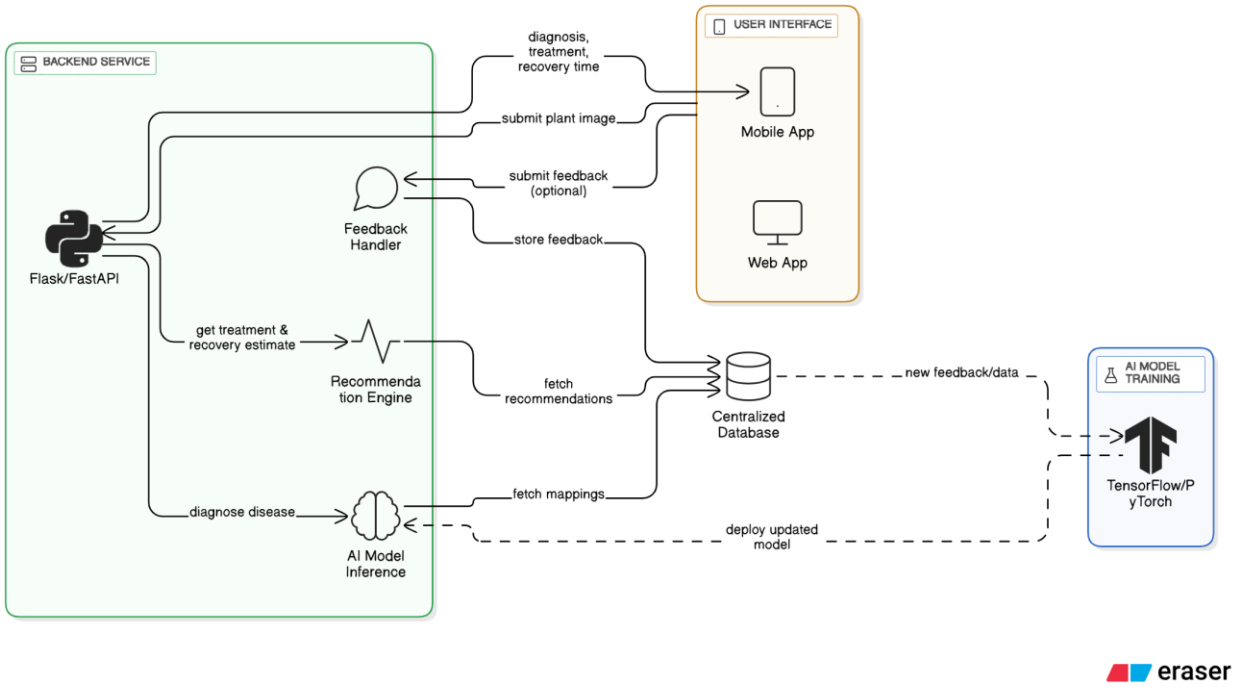


Figure 3: Component Diagram

3.2 Software Methodology

Requirement Gathering and Analysis

- Gather requirements from cultivators and agricultural experts.
- Identify what are the main diseases affect Anthurium .
- Define inputs (plant/flower images) and outputs (disease, treatment, recovery time).

System Design

- Architecture Design:
  - Flutter frontend
  - Python backend with TensorFlow/Keras model.
  - Firebase for storage, authentication, and result logging.

- UI/UX Design: image upload user friendly interface, disease results prompt, treatment recommends UI

Model Development (AI/ML)

- Collect diseased Anthurium plant Anthurium flower images.
- Train deep learning models using Keras/TensorFlow.
- Test accuracy, optimize, and convert to TensorFlow Lite for mobile deployment.

Implementation

- Integrate the trained ML model into the mobile app backend
- Implement recovery time logic.
- Connect Flutter frontend .

Testing

- Test disease detection model with labeled images.
- Integration Testing
- Cultivators test app usability and correctness by their own.

Deployment & Maintenance

- Deploy the mobile app on both the Android Play Store and IOS app stores.
- Improve disease detection.

3.3 Tools and Technologies

1. Flutter -Frontend

- Build the mobile app for both IOS and Android.
- Design user friendly interface and easy navigation for upload photo, treatment recommendations, recovery time
- Integrate with the other component.
- Display disease results and recovery progress .

2. Python -Backend Processing

- Use Python for image preprocessing .
- Develop trained deep learning models for disease detestation.
- Flask or Fast API backend to the Flutter app.

3. TensorFlow & Keras

- Train deep learning image classification models like ResNet, MobileNet
- Use Karas API to develop the disease detection model.
- TensorFlow Lite (TFLite) is using to create a lightweight model directly **.**

4. Firebase -Cloud Services

- allow cultivators to create accounts.
- Cloud Storage use to store uploaded plant/flower images for analysis.
- By using Firestore Database save detection results, treatments, and recovery history.
- In notifications remind cultivators to follow-up treatments and recovery tracking.

3.4 Data Set

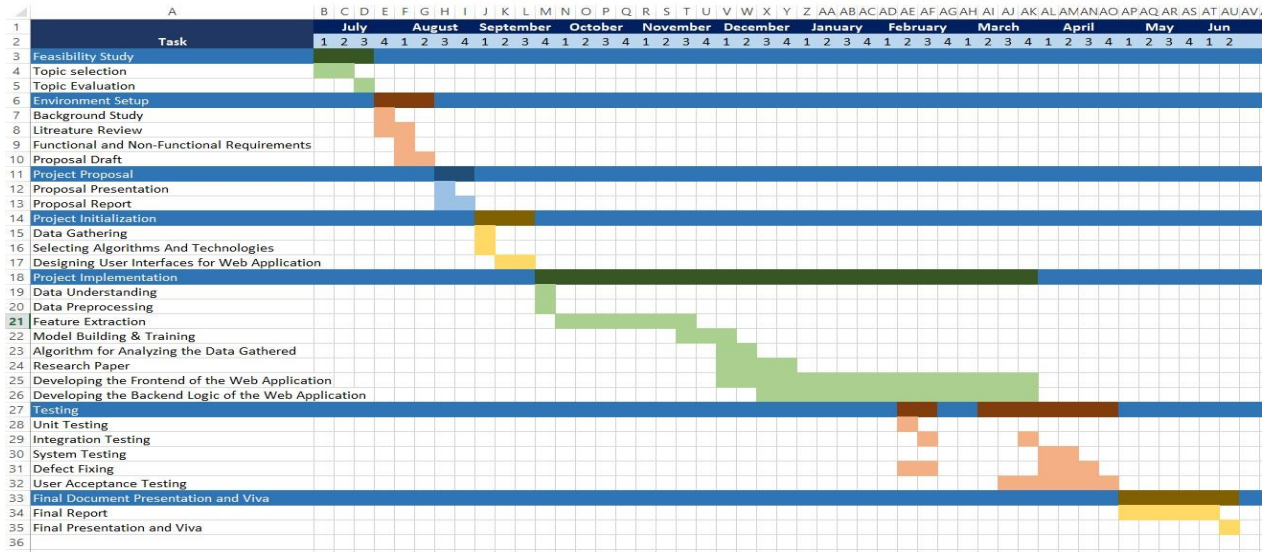
- Collection of diseases about Anthurium

  Anthurium diseases data set collected from Department of agriculture and resources in internet.

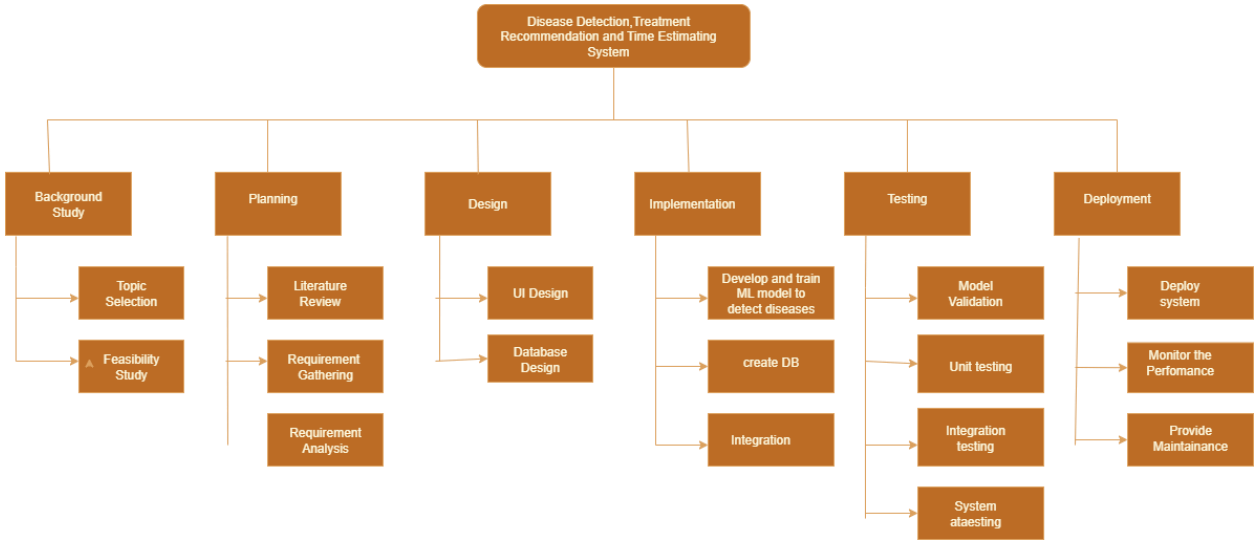
- Diseased Anthurium leaves and flowers

  Hope to find several diseased Anthurium plants and flowers from nearest Anthurium Farming houses.

3.5 Gantt Chart



3.6 Work breakdown chart



4. Project Requirement

4.1 functional Requirements

- User can upload image of Anthurium plant or a flower to the application.

- System validates varies file formate like jpg,png and sizes of images that user uploaded.
- Detect diseases of analyzed pictures by trained deep learning models.
- Estimate recovery time period for diseased plants.
- Display all the details on UI (disease , recovery time etc)
- Backend and frontend communication via Rest API.

4.2 User Requirements

- User friendly interface
- Easy navigation system
- Accessibility
- Data Security
- Performance

4.3 System Requirements

- Flutter
- Python
- TensorFlow
- Flask
- Google firestore

4.3 Non Functional Requirement

- Usability
- Performance
- Security
- Scalability
- Maitane
- Availability

5. Description of personal and Facilities

Member	Component	Task
De Silva W.S.A	Smart detection and treatment estimation of Anthurium diseases using image based AI models.	Develop algorithm to extract data . Create deep learning algorithm to detect diseases of uploaded images. Build a system to identify diseases among data set using machine learning. Collaborated with the group to testing , and accuracy of the system Design user friendly UI and easy navigation system .

6. Budget and Budget Justification

For this research component, we do not aim to commercialize our individual components and the whole research will be commercialized as an application. Following is the budget justification for the whole research system.

Description	**Cost**

Cost for Deployment	10,000.00
Internet	4000.00
Travelling cost for data collection	4000.00
Google Map API	2000.00
Paper Publishing	20,000.00
Total	40,000.00

7.Commercialize

Target Audience

- Commercial Anthurium cultivators / growers
- Home gardeners
- Agricultural Extension Officer
- Researchers
- Plant Pathologists
- Agri-Tech Companies
- Government Agencies

Market Opportunity

- Growing Global Ornamental Plant Market
- Gap in Current Solutions
- High Disease Vulnerability

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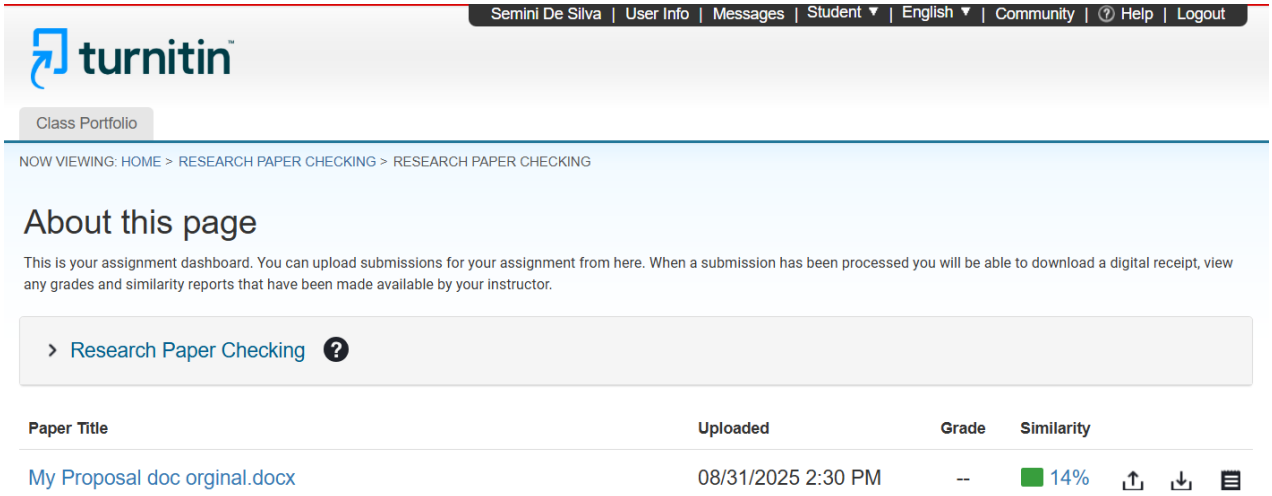
9.Appendices